

Theoretical Foundations of the Wedge-Sharing Constant g_{share} in Emergent Gravity

Section 1: Introduction: The Geometric Origin of Gravitational Coupling

1.1. Context: Gravity from Quantum Information

The quest to unify general relativity and quantum mechanics represents one of the most profound challenges in modern theoretical physics. While conventional approaches have focused on quantizing the gravitational field, treating it as a fundamental interaction mediated by a force-carrying particle (the graviton), a parallel and increasingly influential paradigm posits that gravity is not fundamental at all. Instead, it may be an emergent, macroscopic phenomenon arising from the collective behavior of more fundamental, microscopic degrees of freedom, much as thermodynamics emerges from the statistical mechanics of atoms. This “emergent gravity” perspective recasts spacetime itself as a construct derived from the entanglement and information content of an underlying quantum system.

Historical precedents for this line of inquiry are well-established. In 1967, Andrei Sakharov proposed that gravity could be an “induced” effect, arising from the vacuum fluctuations of quantum matter fields. In 1995, Ted Jacobson demonstrated that Einstein’s field equations could be derived as an equation of state from the thermodynamic properties of local Rindler horizons, suggesting a deep connection between gravity, entropy, and information. More recently, Erik Verlinde’s theory of entropic gravity proposed that gravity is an entropic force, driven by changes in the information associated with the positions of material bodies. A common thread running through these and other emergent gravity frameworks, including those based on tensor networks and Group Field Theory (GFT), is that they often succeed in establishing a proportionality between geometric quantities (like curvature) and physical quantities (like energy-momentum). However, they frequently leave a crucial piece of the puzzle undetermined: the constant of proportionality, which is ultimately related to Newton’s gravitational constant, G . These models typically yield a relationship of the form $G \propto 1/\mathcal{N}$, where $\mathcal{N}$ is some measure of the underlying degrees of freedom, often requiring the imposition of a physical cutoff scale to fix its value.

1.2. The Wedge-Sharing Hypothesis and the Role of g_{share}

The theoretical framework under investigation proposes a novel solution to this long-standing problem. It posits that gravity emerges from the dynamics of entanglement transport, and that the strength of the resulting gravitational interaction is fixed not by an arbitrary cutoff, but by a single, dimensionless, universal constant of geometric origin, denoted g_{share} . This constant is hypothesized to arise from a fundamental capacity constraint on the sharing of information channels.

The physical picture is as follows: spacetime is modeled as a foliation of concentric spherical shells. Entanglement between degrees of freedom on adjacent shells is mediated by a dense network of discrete links. A crucial physical assumption is that these links are subject to a “one-to-one” capacity constraint, meaning each outgoing link from a point on one shell must connect to a unique point on the next. This constraint implies that the entanglement “rays” emanating from any point must share the available 4π steradians of solid angle. This phenomenon is termed the “wedge-sharing” effect, where each link occupies a small solid-angle “wedge.” The collective effect of these links, each undergoing small angular deflections as they propagate from shell to shell, gives rise to the macroscopic curvature of spacetime.

In this model, the constant g_{share} quantifies the average transverse (angular) variance accumulated by these entanglement links per unit of radial propagation, under conditions of near-uniform link density. It is a measure of how efficiently entanglement spreads across the

spherical layers. The framework’s key prediction is that once this single geometric constant is determined, Newton’s constant G can be derived from first principles without any direct gravitational data input. The model’s internal simulations suggest a value of $g_{\text{share}} \approx 7.4$. The fact that this value is of order unity—neither hierarchically large nor small—is significant. It suggests a natural geometric origin, rather than a value requiring fine-tuning or depending on the intricacies of an unknown microscopic theory. It points toward a universal statistical property of random geometric networks on a sphere.

1.3. Report Objectives and Methodological Approach

The primary objective of this report is to provide an independent, *ab initio* derivation or a set of rigorous, analytically-derived bounds for the wedge-sharing constant, g_{share} . The validation sought must be external to the emergent gravity framework in which it was proposed, rooting the constant in established and well-understood mathematical and physical theories. To achieve this, the report pursues two parallel and complementary lines of inquiry.

First, a series of rigorous mathematical derivations will be undertaken. We will model the physical picture of entanglement transport between spherical shells using three distinct but related mathematical formalisms:

- **Optimal Transport on the 2-Sphere (S^2):** The one-to-one matching of entanglement links will be framed as a bipartite optimal transport problem. We will leverage powerful recent results in this field that provide exact asymptotic constants for the expected transport cost.
- **Random Geometric Graphs (RGGs):** The network of links will be modeled as a layered RGG. We will use spectral graph theory to analyze the diffusion of information across the network, relating the graph’s expansion properties to an effective angular variance per layer.
- **Spherical Point Process Statistics:** We will use the tools of stochastic geometry, specifically the statistics of Poisson point processes on the sphere, to establish a baseline for the angular separation between links, providing a rigorous lower bound on the matching variance.

Second, a comprehensive comparative analysis will be conducted to situate g_{share} within the landscape of modern quantum gravity research. We will survey the literature of leading background-independent quantum gravity theories—namely Group Field Theory (GFT), Loop Quantum Gravity (LQG), and Random Tensor Networks (RTNs)—for any analogous universal constants that govern transport, diffusion, entanglement capacity, or the scaling of geometric properties.

By combining direct derivation with comparative analysis, this report aims to either confirm the value of $g_{\text{share}} \approx 7.4$ from first principles or, if a discrepancy is found, to precisely identify the physical or mathematical assumptions that account for the difference. The ultimate goal is to provide a solid theoretical foundation for the wedge-sharing constant, thereby strengthening or refining the emergent gravity framework it underpins.

Section 2: Mathematical Frameworks for Entanglement Transport on the Sphere

To rigorously analyze the wedge-sharing constant, it is necessary to first establish the mathematical formalisms that can model the physical process of capacity-constrained entanglement transport between spherical surfaces. This section introduces three such frameworks: optimal transport theory, which provides a language for describing the most efficient matching between

two distributions; spectral graph theory, which quantifies the diffusive properties of networks; and stochastic geometry, which describes the statistical properties of random point distributions.

2.1. Optimal Transport and the Wasserstein Metric on S^2

The problem of finding the most efficient way to rearrange a distribution of mass from one configuration to another is the central question of optimal transport theory. Originally formulated by Gaspard Monge in 1781, the problem sought a map T that transports a mass distribution μ to a distribution ν while minimizing the total work done, where the work is the integral of the cost of moving a unit of mass from a point x to its destination $T(x)$.

A modern, more flexible formulation was provided by Leonid Kantorovich. Instead of seeking a deterministic map, the Kantorovich problem seeks a “transport plan” π , which is a joint probability distribution on the product space of the source and target domains. The plan $\pi(x, y)$ specifies the amount of mass transported from location x to location y . The set of all valid transport plans, denoted $\Pi(\mu, \nu)$, consists of all joint distributions whose marginals are μ and ν .

For the purposes of this report, the most relevant formulation is for the quadratic cost function, $c(x, y) = d(x, y)^2$, where $d(x, y)$ is the metric distance between points x and y . This cost function gives rise to the squared 2-Wasserstein distance, defined as:

$$W_2^2(\mu, \nu) = \min_{\pi \in \Pi(\mu, \nu)} \int_{X \times X} d(x, y)^2 d\pi(x, y).$$

This quantity represents the minimum possible expected squared displacement under any valid transport plan.

In the context of the user’s model, the physical space is the 2-sphere, S^2 . The appropriate distance metric is the geodesic distance, i.e. the great-circle distance between two points on the sphere. The problem of finding a one-to-one matching between a set of n entanglement endpoints on an inner shell (represented by an empirical measure μ_n) and n endpoints on an outer shell (represented by ν_n) is a discrete bipartite optimal transport problem. The solution to this problem minimizes the sum of squared geodesic distances between matched pairs, which is precisely the discrete version of the W_2^2 distance. This framework thus provides a direct and powerful tool for calculating the expected angular variance that is central to the definition of g_{share} .

2.2. Spectral Properties of Random Geometric Graphs (RGGs)

An alternative way to model the network of entanglement links is as a Random Geometric Graph (RGG). An RGG is constructed by distributing a set of vertices randomly in a metric space and connecting pairs of vertices with an edge if their distance is less than a certain threshold. In our case, the vertices are the entanglement endpoints on the spherical shells, and the “edges” are the one-to-one connections between adjacent shells.

The connectivity and transport properties of a graph can be characterized by the spectrum of its associated graph Laplacian or adjacency matrix. The graph Laplacian L is an operator analogous to the continuous Laplace–Beltrami operator on a manifold. Its spectrum (the set of eigenvalues) encodes crucial information about the graph’s structure. The smallest eigenvalue is always zero, corresponding to a constant function over the vertices. The second-smallest eigenvalue, often denoted λ_2 (or equivalently the second-largest eigenvalue of the adjacency matrix), is of particular importance. The magnitude of λ_2 determines the spectral gap of the graph, which quantifies how well-connected the graph is.

A large spectral gap is the defining feature of an expander graph—a graph that is sparse yet highly connected. The spectral gap is fundamentally related to the mixing time of a random walk on the graph. The mixing time is the number of steps a random walker must take before its position is approximately uniformly distributed over the vertices, regardless of its starting point. A larger spectral gap implies a shorter mixing time, meaning that information or influence diffuses rapidly across the network. By modeling the propagation of entanglement through the layered shells as a random walk on an RGG, we can use the spectral gap to estimate the effective diffusion constant of this process. This diffusion constant is directly related to the per-layer angular variance, providing an independent, graph-theoretic route to constraining g_{share} .

2.3. Stochastic Geometry of Spherical Poisson Point Processes

A third, complementary approach is provided by the field of stochastic geometry, specifically the theory of point processes. To establish a baseline for the angular separation between entanglement links, we can model their endpoints on a spherical shell as a homogeneous Poisson point process. This process is characterized by two key properties: (1) the number of points in any region follows a Poisson distribution with a mean proportional to the area of the region, and (2) the numbers of points in disjoint regions are independent random variables. This model is the mathematical idealization of a “completely random” and uniform distribution of points.

Within this framework, one can analytically compute the distribution of the distance from an arbitrary point to its nearest neighbor in the process. For a Poisson process on a sphere of radius R with intensity λ (points per unit area), the cumulative distribution function (CDF) for the angular distance θ_{NN} to the nearest neighbor is given by:

$$P(\theta_{\text{NN}} \leq \theta) = 1 - \exp[-\lambda A_{\text{cap}}(\theta)],$$

where $A_{\text{cap}}(\theta) = 2\pi R^2(1 - \cos \theta)$ is the area of a spherical cap of angular radius θ .

This formula provides a powerful tool. By differentiating the CDF to find the probability density function (PDF), we can compute the expectation value of any function of the nearest-neighbor angle, such as $\mathbb{E}[\theta_{\text{NN}}^2]$ or $\mathbb{E}[\sin^2 \theta_{\text{NN}}]$. This calculation serves as a fundamental baseline. A simple “greedy” matching algorithm would pair each point with its nearest available neighbor. While not globally optimal, the statistics of this greedy matching are closely related to the nearest-neighbor distribution. As will be shown, this allows us to establish a rigorous lower bound on the variance of any optimal one-to-one matching, providing a crucial piece of the puzzle for bounding g_{share} .

Section 3: Derivation from Optimal Transport and Bipartite Matching on S^2

The framework of optimal transport provides the most direct and rigorous avenue for calculating the per-layer angular variance that defines g_{share} . By modeling the one-to-one mapping of entanglement links between successive spherical shells as a bipartite matching problem, we can leverage a powerful and definitive result from the mathematical literature to derive the precise asymptotic value of the relevant geometric constant.

3.1. The Asymptotic Cost of 2D Matching: The Ambrosio–Stra–Trevisan Limit

The problem of determining the expected cost of optimally matching two sets of random points has a long history, with a celebrated result by Ajtai, Komlós, and Tusnády (AKT) establishing the correct scaling law in two dimensions. For the quadratic cost problem, the expected total cost was known to scale as $\sim \frac{\ln n}{n}$. However, the precise value of the leading coefficient remained a conjecture for many years, particularly one arising from a PDE-based ansatz in the physics literature.

This conjecture was rigorously proven in a landmark 2016 paper by Luigi Ambrosio, Federico Stra, and Dario Trevisan. Their main theorem provides the exact leading term in the asymptotic expansion of the expected quadratic transportation cost for two-dimensional random matching problems.

The key result, as it applies to the bipartite matching problem, is as follows: Let D be a compact 2-dimensional Riemannian manifold without boundary and with unit volume (e.g., a 2-sphere of area 1). Let $\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_i}$ and $\nu_n = \frac{1}{n} \sum_{i=1}^n \delta_{Y_i}$ be two empirical measures formed from n independent and identically distributed random points X_i, Y_i drawn from the uniform measure on D . Then the expected squared 2-Wasserstein distance between them has the asymptotic limit:

$$\lim_{n \rightarrow \infty} \frac{n}{\ln n} E = \frac{1}{2\pi}.$$

This result is exceptionally robust; the authors note that the geometry of the domain enters the proof only through the asymptotic properties of the spectrum of its Laplacian operator, making the result applicable to any compact 2D manifold, including the sphere.

3.2. Mapping the Mathematical Result to the Physical Model

To apply this powerful mathematical result to the physical model of emergent gravity, we must carefully map the corresponding quantities:

- **The Matching Problem:** The user's model of a one-to-one matching of entanglement links between two adjacent shells with n endpoints each is a direct physical realization of the bipartite matching problem between the empirical measures μ_n and ν_n .
- **The Cost Function:** The quantity E represents the expected total cost of the optimal matching, where the cost is the sum of the squared geodesic distances between matched pairs.
- **Average Per-Point Cost:** The average squared geodesic distance per matched pair is $\langle d^2 \rangle = E/n$.
- **Angular Variance:** On a sphere of radius R , the geodesic distance d and the angular separation θ are related by $d = R\theta$. Consequently, the average squared angular separation is $\langle \theta^2 \rangle = \langle d^2 \rangle / R^2$.

Combining these points, we can use the Ambrosio–Stra–Trevisan (AST) limit to find the asymptotic behavior of the average squared angular separation in the matching. Rearranging the AST result gives:

$$E \approx \frac{\ln n}{n} \cdot \frac{1}{2\pi}.$$

This formula is valid for a manifold of unit area. For a sphere of radius R , the area is $A = 4\pi R^2$. To apply the theorem, we can consider a rescaled sphere of unit area, which corresponds to setting $R = 1/\sqrt{4\pi}$. On this unit-area sphere, the average squared geodesic distance per pair is:

$$\langle d^2 \rangle = \frac{1}{n} E \approx \frac{\ln n}{n^2} \cdot \frac{1}{2\pi}.$$

The corresponding average squared angular separation on the original sphere of radius R is:

$$\langle \theta^2 \rangle = \frac{\langle d^2 \rangle}{R^2} = \frac{1}{R^2} \cdot \frac{\ln n}{n^2} \cdot \frac{1}{2\pi}.$$

This expression gives the full dependence of the average squared angle on the number of points n and the radius of the sphere R .

3.3. Calculating the Per-Layer Angular Variance (b^2)

The user’s framework seeks a local, density-dependent constant, b^2 , representing the per-layer variance. The prompt suggests that the relevant quantity to extract is the constant prefactor in the asymptotic scaling relation, with the identification $g_{\text{share}} = b^2$. The AST result provides exactly this prefactor.

The formula $E \sim \frac{\ln n}{n} \cdot C$ shows that the constant of proportionality is $C = 1/(2\pi)$. This constant is dimensionless and universal for all 2D manifolds. In the context of the user’s model, this constant is the quantity sought. Therefore, based on the most rigorous and directly applicable result in the mathematical literature on optimal transport, the per-layer angular variance constant is:

$$b^2 = \frac{1}{2\pi} \approx 0.15915.$$

3.4. From Per-Layer Variance to the Cumulative Constant g_{share}

According to the user’s problem statement, the wedge-sharing constant g_{share} is identified directly with this per-layer variance constant, i.e. $g_{\text{share}} = b^2$. This identification implies that the emergent gravity framework’s central geometric parameter should be:

$$g_{\text{share}} = \frac{1}{2\pi} \approx 0.159.$$

This result is derived directly from a rigorously proven theorem in optimal transport theory, under the sole assumption that the physical process of entanglement matching is correctly modeled by a bipartite optimal matching with a quadratic cost in geodesic distance.

This analytical derivation leads to a significant conclusion: the value of the constant derived from first principles in mathematics (~ 0.159) is in stark disagreement with the value of ~ 7.4 obtained from the user’s internal simulations. The derived value is smaller by a factor of approximately 46. This discrepancy is a central finding of this report and indicates that there must be a deeper physical or structural assumption within the user’s emergent gravity framework that is not captured by the standard mathematical formulation of the optimal matching problem. This critical point will be revisited and analyzed in detail in Section 7.

Table 1: Asymptotic Constants in 2D Optimal Transport.

Problem Type	Cost Function	Asymptotic Formula for E	Constant	Reference
Bipartite Matching (2D manifold)	$d(x, y)^2$	$\lim_{n \rightarrow \infty} \frac{n}{\ln n} E$	$\frac{1}{2\pi}$	Ambrosio–Stra–Trevisan (2016)
Matching to Reference Measure (2D)	$d(x, y)^2$	$\lim_{n \rightarrow \infty} \frac{n}{\ln n} E$	$\frac{1}{4\pi}$	Ambrosio–Stra–Trevisan (2016)
Bipartite Matching (Planar)	$d(x, y)^2$	$E \sim C \frac{\ln n}{n}$	$C = \frac{1}{2\pi}$	AKT (1984); Caracciolo et al. (2014)

Section 4: Bounds from Random Geometric Graph Diffusion and Spectral Analysis

An independent line of inquiry into the nature of g_{share} can be pursued by modeling the entanglement network not as an optimal flow problem, but as a medium for diffusion. By representing the layered network of one-to-one connections as a Random Geometric Graph, we can use the tools of spectral graph theory to analyze how information spreads angularly. This approach provides a complementary perspective to optimal transport, focusing on the statistical properties of random walks rather than the minimization of a global cost function.

4.1. The Layered Network as a Random Geometric Graph

We model the entanglement network as a sequence of layers, where each layer consists of n vertices (entanglement endpoints) distributed uniformly on a sphere. The connections between layer i and layer $i+1$ form a perfect matching. While this is not a standard RGG (where connections are based on proximity), in a system with high density and local connections, the properties are closely related. The fundamental question remains the same: what is the characteristic angular displacement for a single step from one layer to the next? This displacement can be viewed as one step of a random walk that moves from vertex to vertex across the layers. The variance of this angular step is precisely the quantity $b^2 = g_{\text{share}}$.

4.2. The Spectral Gap and Angular Mixing

The rate at which a random walk on a graph converges to its stationary distribution (i.e. how quickly it “forgets” its starting point) is governed by the graph’s spectral gap. For a regular graph, the spectral gap is given by $1 - \lambda_2$, where λ_2 is the second-largest eigenvalue of the graph’s normalized adjacency matrix. A larger spectral gap implies faster mixing and more rapid diffusion.

Recent work on RGGs in high-dimensional spheres S^{d-1} has provided bounds on this second eigenvalue. Specifically, theoretical results indicate that with high probability, $\lambda_2 \approx \tau$ under a wide range of parameters, for an RGG where edges are formed between points with an inner product greater than a threshold τ . Since the inner product between two unit vectors is $\cos \theta$, the connection threshold τ corresponds to a maximum connection angle $\theta_{\text{max}} = \arccos(\tau)$.

This result connects the macroscopic spectral properties of the graph to the microscopic geometric rule for forming connections. A smaller connection radius (larger τ) leads to a sparser graph. Intuitively, a random walk on a graph with only very local connections will diffuse slowly, corresponding to a small angular variance per step. Conversely, a graph with long-range connections will mix rapidly, leading to a larger variance.

4.3. An Upper Bound on g_{share} from Graph Expansion

While the RGG literature does not provide a direct formula for the per-step variance of a layered matching, it provides the tools to bound it. The spectral gap sets a limit on how “non-random” a single step can be. A graph that is a good expander (large spectral gap, small λ_2) cannot have bottlenecks that trap a random walker. This means that the probability of taking a very small angular step is limited; the walk must spread out.

The connection between the optimal flow problem of Section 3 and the diffusion problem considered here is profound. The optimal transport map describes the most efficient “geodesic” flow of mass, while a random walk describes the statistical average of all possible paths. The heat kernel, which was central to the PDE-based proof of the AST limit, is the propagator for a diffusion process (Brownian motion) on the manifold. The AST result, which gives an exact constant for the optimal transport cost, can be seen as the continuum limit of a discrete matching process. Similarly, the diffusion constant of a random walk on a sequence of finer and finer RGGs should converge to the diffusion constant of Brownian motion on the sphere.

Therefore, the two approaches must be consistent. The exact constant $b^2 = 1/(2\pi)$ derived from optimal transport theory should be recoverable, in principle, from a spectral analysis of the corresponding RGG in the continuum limit. While the current RGG literature may not be developed enough to extract this constant with the same precision, it provides a powerful conceptual validation. The analysis confirms that a universal, density-dependent constant governing angular diffusion is expected to emerge from the network’s statistical properties. The

spectral analysis provides a framework for understanding how this constant depends on the local connectivity rules of the entanglement network, reinforcing the conclusion that a constant like g_{share} should exist, even if its precise value remains a point of contention.

Section 5: Analysis via Spherical Point Process Statistics

A third method for constraining g_{share} comes from the direct statistical analysis of point distributions on the sphere. By modeling the entanglement endpoints as a random point process, we can calculate a baseline for the typical angular separation between links. This approach, rooted in stochastic geometry, allows for the establishment of a rigorous lower bound on the matching variance and provides a complementary, non-asymptotic perspective to the optimal transport analysis.

5.1. Baseline Calculation: Expected Nearest-Neighbor Angle

As a foundational model, we consider the endpoints of the entanglement links on a spherical shell to be distributed according to a homogeneous Poisson point process with intensity λ (points per unit area). The probability that the nearest neighbor to a given point lies within an angular distance θ is determined by the probability that a spherical cap of that angular radius is not empty. This leads to the cumulative distribution function (CDF):

$$F(\theta) = P(\theta_{\text{NN}} \leq \theta) = 1 - \exp[-\lambda A_{\text{cap}}(\theta)].$$

The corresponding probability density function (PDF) is obtained by differentiation:

$$p(\theta) = \frac{dF(\theta)}{d\theta} = 2\pi\lambda R^2 \sin \theta \exp[-\lambda A_{\text{cap}}(\theta)].$$

From this PDF, we can compute the expected value of any function of the nearest-neighbor angle. A quantity of interest, related to the squared displacement for small angles, is the expected value of $\sin^2 \theta$. The expectation is given by the integral:

$$\mathbb{E}[\sin^2 \theta_{\text{NN}}] = \int_0^\pi \sin^2 \theta p(\theta) d\theta = 2\pi\lambda R^2 \int_0^\pi \sin^3 \theta \exp[-\lambda A_{\text{cap}}(\theta)] d\theta.$$

This integral can be evaluated analytically. For the case of unit intensity on a sphere scaled to have an area of 4π (i.e. $\lambda R^2 = 1$), the result is:

$$\mathbb{E}[\sin^2 \theta_{\text{NN}}] = \frac{(-1 + 2\pi) + (1 + 2\pi)e^{-4\pi}}{2\pi^2} \approx 0.26765.$$

This value represents the average squared sine of the angular separation to the closest point in a completely random distribution. It serves as a hard, analytically-derived baseline.

5.2. The Gap Between Nearest-Neighbor and Optimal Matching Distances

The nearest-neighbor (NN) matching strategy is a “greedy” algorithm: it pairs each point with its closest available partner sequentially, without regard for the global consequences of that choice. In contrast, an optimal matching algorithm seeks to minimize the total cost over all pairs. This global optimization constraint means that a point may be forced to match with a partner that is not its nearest neighbor, because its nearest neighbor is the optimal partner for another point.

This fundamental difference implies that the average matching distance in an optimal pairing must be greater than or equal to the average distance in a pure nearest-neighbor scenario (where multiple points could claim the same nearest neighbor). Therefore, the expected squared

displacement in an optimal matching is bounded from below by the expected squared nearest-neighbor distance:

$$\mathbb{E}[\theta_{\text{match}}^2] \geq \mathbb{E}[\theta_{\text{NN}}^2].$$

Using the small-angle approximation $\sin \theta \approx \theta$, we can employ the result from the previous section to establish a rigorous lower bound on the per-layer variance constant b^2 :

$$g_{\text{share}} = b^2 \gtrsim \mathbb{E}[\sin^2 \theta_{\text{NN}}] \approx 0.26765.$$

This provides a firm, non-asymptotic floor for the value of the wedge-sharing constant, derived from first principles of geometric probability.

5.3. Applying Corrections from Gravitational Allocation Models

While the NN calculation provides a lower bound, we also need to constrain how much larger the optimal matching distance can be. The “gravitational allocation” model developed by Holden, Peres, and Zhai provides a powerful tool for this purpose. Their method defines a near-optimal matching by partitioning the sphere into cells of equal area, one for each point, based on the flow lines of a potential field analogous to Newtonian gravity. Each point is then matched to the corresponding point in the other set’s partition.

The significance of this work is that it proves that the expected distance for their matching scheme scales as $O(\sqrt{\ln n})$ for a sphere of area n , which is the same optimal scaling rate as the true minimal matching. This demonstrates that their constructive allocation method is within a constant factor of the true optimum. The diameter of the cells in their allocation is of the same order as the nearest-neighbor distance. Since a point and its match lie within corresponding cells, the matching distance is also constrained to be of the order of the nearest-neighbor distance.

While the exact ratio between the expected optimal matching distance and the expected nearest-neighbor distance is not known, the gravitational allocation result strongly suggests that this ratio is a constant of order unity. It shows that the cost of enforcing the global one-to-one constraint does not cause the average matching distance to diverge significantly from the natural local scale set by the nearest-neighbor distance.

5.4. Establishing a Rigorous Bound on g_{share}

By combining the insights from the Poisson baseline and the gravitational allocation model, we can establish a credible range for g_{share} :

- **Lower Bound:** As rigorously derived, $g_{\text{share}} = b^2$ must be greater than the baseline variance set by the nearest-neighbor distribution. Normalizing for unit density, we have the firm bound $g_{\text{share}} > 0.267$.
- **Upper Bound:** While a rigorous upper bound is more elusive, the gravitational allocation results suggest that the correction factor is not large. Empirical studies of matching algorithms often show that greedy methods perform nearly as well as optimal ones when competition for matches is low, which is the case for a uniform distribution. It is physically reasonable to assume that the average squared matching distance will not be more than, for example, a factor of 4 or 5 larger than the nearest-neighbor baseline. This would suggest a heuristic upper bound of $g_{\text{share}} < 1.5$.

This analysis from stochastic geometry thus brackets the wedge-sharing constant in the approximate range $0.27 < g_{\text{share}} < 1.5$. This result is significant for two reasons. First, it independently confirms that a geometric constant of order unity is expected to arise from such a matching problem. Second, this range is also in stark contradiction with the user’s simulated

value of ≈ 7.4 . This reinforces the conclusion from the optimal transport analysis: the standard mathematical models of random one-to-one matching on a sphere do not produce a constant of this magnitude.

Section 6: Comparative Analysis of Constants in Quantum Gravity Models

To place the wedge-sharing constant g_{share} in the broader context of modern physics, it is essential to compare it with analogous universal constants that appear in other approaches to quantum gravity. This section surveys the literature on Group Field Theory (GFT), spin foams, Loop Quantum Gravity (LQG), and Random Tensor Networks (RTNs) to identify parameters that, like g_{share} , govern the transport of information, the flow of entanglement, or the effective dimensionality of emergent spacetime.

6.1. Transport and Diffusion in Group Field Theory and Spin Foams

Group Field Theory is a quantum field theory approach to quantum gravity where the fundamental fields create and annihilate “atoms” of space, represented as spin network vertices or simplices. The collective quantum dynamics of these fields is intended to describe the emergence of spacetime. One of the key observables used to characterize the geometry of this emergent spacetime is the spectral dimension, D_s .

The spectral dimension is not the usual topological dimension but rather the effective dimension experienced by a diffusing random walker. It is defined through the return probability of a diffusion process, $P_{\text{return}}(T) \sim T^{-D_s/2}$, where T is the diffusion time. A remarkable and robust feature found in several quantum gravity approaches, including GFT, spin foams, and Causal Dynamical Triangulations (CDT), is that the spectral dimension is scale-dependent, or “running.” A common result is a dimensional flow: at very small scales (probing the Planck regime), spacetime appears two-dimensional ($D_s \approx 2$), while at large scales (the classical regime), it flows to the familiar four dimensions ($D_s = 4$).

This phenomenon of a running spectral dimension provides a compelling conceptual link to the user’s framework. The constant g_{share} can be interpreted as a parameter that characterizes the macroscopic diffusion process in the $D_s = 4$ regime. The fact that GFT and related models independently predict an emergent 4D spacetime at large scales is a point of deep consistency. However, the GFT literature typically focuses on calculating the value of the spectral dimension itself, rather than the specific value of the diffusion constant in a given dimensional regime. While these theories predict that a diffusion process exists, they have not, to date, provided an *ab initio* calculation of a universal constant analogous to g_{share} . The running of the dimension is often tied to the renormalization group flow of the theory’s coupling constants, suggesting that a future, more complete understanding of the GFT fixed point could potentially determine such a constant.

6.2. Entropy Prefactors in Loop Quantum Gravity

Loop Quantum Gravity (LQG) is a canonical approach to quantum gravity where the quantum states of geometry are described by spin networks—graphs with edges labeled by representations of $\text{SU}(2)$ and vertices labeled by intertwiners. A central result in quantum gravity, inspired by black hole thermodynamics and the holographic principle, is the Bekenstein–Hawking area law for entropy: $S = A/(4G_N\hbar)$. The dimensionless prefactor of $1/4$ is a crucial benchmark that any successful theory of quantum gravity must reproduce.

In the context of LQG, entanglement entropy calculations also yield an area law. When a region of space is partitioned, the entanglement between the inside and outside is proportional to the area of the boundary surface. Significant effort has been devoted to deriving the $1/4$

prefactor from the microstate counting of the spin network degrees of freedom that puncture the surface.

For the user’s emergent gravity framework, this provides a powerful consistency check. The framework must ultimately be able to reproduce the area law with the correct prefactor. This implies a relationship between the fundamental constants of the model, namely g_{share} , and the observed macroscopic constants like G_N . A successful derivation would likely take the form $1/(4G_N) \propto f(g_{\text{share}})$, where f is a function determined by the structure of the effective field theory. This connection serves as a target for the completed theory, but it does not provide an independent means of deriving g_{share} , as it requires G_N (Newton’s constant) as an input.

6.3. Entanglement Capacity and Flow in Random Tensor Networks

Random Tensor Networks (RTNs) offer a concrete and computable toy model for holography and emergent geometry. In these models, the entanglement entropy of a boundary region is given by the max-flow/min-cut theorem: it is proportional to the minimum number of network links one must cut to separate the region from its complement, a direct analogue of the Ryu–Takayanagi formula.

For tensors with a large bond dimension D (representing a large number of underlying degrees of freedom), the entanglement entropy saturates this bound with high probability. This implies that the entanglement flow is “geometric” and that each link crossing the minimal surface is at its maximum capacity. This concept is directly analogous to the user’s “one-to-one capacity” constraint, which posits that each entanglement link is a fully utilized channel.

Furthermore, the structure of tensor network renormalization algorithms, such as the Multi-scale Entanglement Renormalization Ansatz (MERA), provides a direct structural parallel to the user’s model of concentric shells. MERA consists of layers of tensors that systematically remove local entanglement to produce a coarse-grained description of the quantum state at a larger scale. The flow of information from the fine-grained “bulk” to the coarse-grained “boundary” is an explicit, computable process. One could, in principle, study the transverse spreading or “fan-out” of information as it flows through the layers of the MERA network. However, the literature on these models has primarily focused on the scaling of entanglement entropy itself (e.g. logarithmic scaling for critical systems) rather than on extracting universal dimensionless constants that govern the geometry of the information flow.

Table 2: Universal Constants and Parameters in Quantum Gravity Frameworks.

Framework	Constant/Parameter	Definition & Context	Value/Behavior	Analogy to g_{share}
GFT / Spin Foams	Spectral Dimension (D_s)	Diffusion: $P(T) \sim T^{-D_s/2}$	$2 \rightarrow 4$ (running)	Diffusion constant in IR
LQG	Entropy Prefactor (κ)	$S = \kappa(A/L_P^2)$	$\kappa = 1/4$	Consistency w/ G_N
RTN / MERA	Bond Dimension (D)	Local dof per link	Model parameter	Capacity constraint analogue
All	Running Couplings	RG flow / fixed points	Theory-dependent	g_{share} as IR value

Section 7: Synthesis and Constraints on the Wedge-Sharing Constant

The preceding analyses have provided a series of independent, analytically-derived constraints on the wedge-sharing constant, g_{share} . The most striking result is the significant and robust discrepancy between the value predicted by standard mathematical models of optimal matching and the value suggested by the user’s internal simulations. This section synthesizes these findings, explores potential resolutions for the discrepancy, and presents a final assessment of the constant’s theoretical status.

7.1. Reconciling the Discrepancy

The investigation has yielded three key numerical results for the per-layer angular variance constant, $b^2 = g_{\text{share}}$:

- **From Optimal Transport (Section 3):** A rigorous, asymptotic derivation based on the work of Ambrosio, Stra, and Trevisan yields an exact value of $b^2 = 1/(2\pi) \approx 0.159$.
- **From Point Process Statistics (Section 5):** An analysis of nearest-neighbor distances in a spherical Poisson point process establishes a rigorous lower bound of $b^2 \gtrsim 0.267$.
- **From the User’s Framework:** Internal simulations suggest a value of $g_{\text{share}} \approx 7.4$.

There is a clear conflict. The two analytical derivations, while differing slightly due to their distinct assumptions (asymptotic optimal matching vs. non-asymptotic nearest-neighbor baseline), both point to a value of order 10^{-1} . This is more than an order of magnitude smaller than the simulated value of 7.4. This discrepancy cannot be attributed to approximation errors; the optimal transport result is asymptotically exact. Therefore, the resolution must lie in the mapping between the simplified mathematical problem and the more complex physics of the user’s emergent gravity model. We explore two primary hypotheses for the origin of this difference.

7.2. Hypothesis 1: A Different Cost Function

The derivation in Section 3 was predicated on the assumption that the “cost” of matching two entanglement endpoints is given by the square of the geodesic distance between them, $c(\theta) = (R\theta)^2$. This is the standard choice in the mathematical literature on quadratic optimal transport. However, the physical principle described by the user is “wedge-sharing,” which relates to the allocation of solid angle. It is plausible that this physical principle translates into a different, non-standard effective cost function for the matching problem.

For example, the cost might be more sensitive to large angular deviations than the simple quadratic cost. A cost function that grows more rapidly with angle, such as $c(\theta) \sim \tan^2(\theta/2)$ or another function that heavily penalizes pairings across large angular separations, could potentially lead to a different optimal matching configuration with a larger average displacement. The gravitational allocation model (a constructive near-optimal matching scheme based on a gravitational potential) also suggests that effective costs in such problems can take more complex forms than simple metric distances.

Investigating the asymptotic behavior of the matching problem under such non-standard cost functions is a non-trivial mathematical task and lies beyond the scope of this report. However, this hypothesis represents a concrete and physically motivated avenue for future research. It is plausible that the value $g_{\text{share}} \approx 7.4$ is the correct geometric constant for a matching problem governed by a cost function that more faithfully represents the physics of solid-angle capacity constraints.

7.3. Hypothesis 2: Normalization and Physical Interpretation

A second possibility is that the discrepancy arises not from the cost function, but from the definition of g_{share} within the user’s effective field theory (EFT). The constant $b^2 = 1/(2\pi)$ derived from optimal transport is a pure, dimensionless number that emerges from the geometry of the matching problem itself. It is possible that the macroscopic coupling constant that appears in the EFT, which the user has labeled g_{share} , is not this bare geometric constant but is instead related to it by other fundamental geometric or physical factors.

For instance, the physics might involve normalization by the total solid angle of the sphere, 4π . A hypothetical relationship of the form $g_{\text{share}} = (4\pi)^k \cdot b^2$ for some integer k could bridge

the gap. For example, if $k = 1$, then $g_{\text{share}} = 4\pi \cdot (1/(2\pi)) = 2$. If $k = 2$, then $g_{\text{share}} = (4\pi)^2 \cdot (1/(2\pi)) = 8\pi \approx 25.1$. Neither of these simple powers yields 7.4, but they illustrate the principle. The correct normalization factor, if one exists, would need to be derived from the detailed structure of the EFT and the precise way in which the microscopic angular variance is coarse-grained into the macroscopic gravitational coupling. The name “wedge-sharing” itself suggests that ratios involving 4π are physically relevant. This hypothesis suggests that the user should re-examine the derivation of the macroscopic equations of their model to trace the emergence of all numerical factors.

7.4. Final Bounds and Confidence

Based on the comprehensive analysis performed, this report establishes the following definitive conclusions regarding the wedge-sharing constant:

- **Rigorous Analytical Value:** For the standard mathematical problem of bipartite optimal matching of random points on a 2-sphere with a quadratic geodesic cost, the universal constant governing the per-layer angular variance is precisely $b^2 = 1/(2\pi)$.
- **Rigorous Lower Bound:** The statistics of nearest-neighbor distances in a random point process provide a rigorous lower bound on the constant, of the order $b^2 \gtrsim 0.27$.
- **Confidence Assessment:** The confidence in the analytical value of $1/(2\pi)$ for the simplified problem is very high, as it rests on a proven mathematical theorem. The confidence that this value directly corresponds to the user’s g_{share} is low, given the large discrepancy with the simulated value.

The most likely conclusion is that the user’s model contains additional physical content that is not captured by the standard optimal transport problem. The value $g_{\text{share}} \approx 7.4$ is not derivable from the simplified geometric and statistical models investigated in this report. Its origin must lie in a more nuanced effective cost function or in the specific normalization conventions of the user’s theoretical framework.

Section 8: Conclusion: Implications for the Emergent Gravity Framework

This report has conducted an exhaustive, multi-pronged investigation into the theoretical foundations of the wedge-sharing constant, g_{share} , a central parameter in a novel emergent gravity framework. By applying rigorous methods from optimal transport theory, spectral graph theory, and stochastic geometry, and by performing a comparative analysis with related constants in modern quantum gravity research, we have arrived at a series of definitive conclusions and actionable recommendations.

8.1. Summary of Findings

The central objective was to provide an independent, *ab initio* derivation or bounding of g_{share} . Our findings can be summarized as follows:

- **Optimal Transport Derivation:** The most direct mathematical analogue of the user’s model—bipartite optimal matching on the 2-sphere with a quadratic cost—yields an exact, universal constant for the per-layer angular variance: $g_{\text{share}} = 1/(2\pi) \approx 0.159$. This result is based on the rigorously proven asymptotic limit by Ambrosio, Stra, and Trevisan.
- **Rigorous Bounds:** Analysis of nearest-neighbor statistics for a Poisson point process on the sphere establishes a rigorous lower bound of $g_{\text{share}} \gtrsim 0.27$. Combined with less formal arguments from near-optimal matching schemes, this brackets the constant within the order-unity range, but significantly below the user’s simulated value.

- **Comparative Analysis:** A survey of Group Field Theory, Loop Quantum Gravity, and Random Tensor Networks revealed no pre-existing, calculated constant that directly corresponds to g_{share} . However, these theories provide strong conceptual support for key features of the user’s model, such as the emergence of a 4D macroscopic spacetime with diffusive properties (the running spectral dimension) and the geometric nature of entanglement flow (area laws and max-flow/min-cut theorems).

The primary outcome of this investigation is the discovery of a significant numerical discrepancy between the constant derived from established mathematical theory (~ 0.16) and the value required by the user’s framework (~ 7.4).

8.2. Assessing the Plausibility of $g_{\text{share}} \approx 7.4$

The value $g_{\text{share}} \approx 7.4$ is not, in itself, implausible. As an order-unity dimensionless number, it is precisely the kind of constant one might expect to emerge from a fundamental theory without fine-tuning. However, this report concludes that this value cannot be derived from the standard mathematical frameworks for random geometric matching with a simple squared-distance cost.

The origin of the value 7.4 must therefore lie in specific physical assumptions embedded within the user’s model that distinguish it from these simplified mathematical problems. The discrepancy is not a failure of the model, but rather a crucial clue that points toward the precise features of the theory that are responsible for generating the correct gravitational strength. The physics of “wedge-sharing” must be more complex than a simple minimization of geodesic distance.

8.3. Future Research Directions

This report’s findings illuminate two clear and critical paths for future research to solidify the theoretical foundations of the user’s emergent gravity framework:

- **Refine the Model’s Normalization:** A thorough re-examination of the derivation of the macroscopic field equations from the microscopic entanglement dynamics is required. The goal should be to meticulously track all geometric factors (such as powers of 4π) and statistical averaging factors to determine the precise analytical relationship between the macroscopic constant g_{share} and the microscopic per-layer angular variance, b^2 . It is possible that the relationship is not the simple identity $g_{\text{share}} = b^2$, but rather $g_{\text{share}} = f(b^2)$, where the function f contains the physics of the coarse-graining procedure.
- **Formulate and Analyze the Effective Cost Function:** The physical principle of “wedge-sharing” under a one-to-one capacity constraint should be translated into a precise mathematical cost function, $c(\theta)$, for the bipartite matching problem. This function will likely differ from the standard $c(\theta) \propto \theta^2$. The next step would be to analyze the asymptotic behavior of the optimal transport problem with this new, physically-motivated cost function. This is a challenging mathematical problem but one that goes to the heart of the model’s core hypothesis.

8.4. Concluding Remarks

The emergent gravity framework investigated here is built on a compelling and powerful idea: that the strength of gravity is not a fundamental, input parameter of nature but is instead an output, fixed by the universal geometric and informational properties of quantum entanglement. The model’s reliance on a single, falsifiable constant, g_{share} , is a significant theoretical strength, as it exposes the theory to rigorous mathematical scrutiny.

While this report did not find a direct confirmation of the value $g_{\text{share}} \approx 7.4$ from existing mathematical literature, it has successfully sharpened the question. We have moved from “Where does 7.4 come from?” to “What physical principle, embodied as a non-standard transport cost or a specific normalization in the EFT, leads to a geometric variance of 7.4?”. The analyses herein have provided the precise value for the standard case ($1/(2\pi)$) and established the theoretical tools needed to answer this more refined question. By clarifying the path forward, this investigation represents a critical step in validating and developing a promising new theory of quantum gravity.

Code: gshare_brownian.py (verbatim)

```
# gshare_brownian.py # Estimate g_share from transverse Brownian displacement: g_share = b^2
import numpy as np
from numpy.random import default_rng
from math import sqrt
def estimate_gshare(D_perp=1.85, # transverse diffusion coeff (per unit radius)
                    R_list=(24,32,40,48,56),
                    n_samples=200000, # number of rays per radius (vectorized)
                    seed=2025):
    """
    For each R, sample X,Y ~ N(0, 2 D_perp R) independently (no loops over steps).
    Then  = sqrt(X^2 + Y^2) / R. Report b(R) = _rms * sqrt(R), and g_share = (mean b)^2.
    """
    rng = default_rng(seed)
    b_vals = []
    out = []
    for R in R_list:
        # per-axis variance after distance R
        var = 2.0 * D_perp * R
        # draw endpoints directly
        X = rng.normal(0.0, sqrt(var), size=n_samples)
        Y = rng.normal(0.0, sqrt(var), size=n_samples)
        r_perp = np.sqrt(X*X + Y*Y)
        theta = r_perp / R
        theta_rms = np.sqrt(np.mean(theta*theta))
        b_R = theta_rms * sqrt(R)
        b_vals.append(b_R)
        out.append((R, theta_rms, b_R))
    b_vals = np.array(b_vals, dtype=float)
    b_mean = float(np.mean(b_vals))
    b_std = float(np.std(b_vals, ddof=1))
    g_share = b_mean * b_mean
    return out, b_mean, b_std, g_share
if __name__ == "__main__":
    rows, b_mean, b_std, g_est = estimate_gshare(D_perp=1.85, R_list=(24,32,40,48,56), n_sar
    print("R _rms b(R)=_rmsR")
    for R, th, bR in rows:
        print(f"{R:<4d} {th: .6f} {bR: .6f}")
    print(f"\nMean b  {b_mean:.6f} ± {b_std:.6f}")
    print(f"g_share  b^2  {g_est:.6f} (target  7.4 when D_perp = 7.4/4 = 1.85)")
```

Output (verbatim)

```
R _rms b(R)=_rmsR
24 0.555480 2.721286
32 0.481108 2.721559
40 0.430171 2.720638
48 0.393105 2.723509
56 0.363125 2.717380
Mean b 2.720874 ± 0.002227
g_share b^2 7.403157 (target 7.4 when D_perp = 7.4/4 = 1.85)
```

Code: One-step transport (fixed) (verbatim)

```
# One-step transport (fixed):
# - g0 from ANALYTIC uniform-in-cap expectation (no greedy matching bias, no Monte Carlo noise)
# - estimated spectrally on a k-NN graph (uniform next-shell), then CLIPPED by geometric cap
# - Scan across [1°, 90°] and solve for that hits a target g (default 7.4)
# Paste this single cell into Colab and run.
import math, json, numpy as np, sys, subprocess
# --- SciPy (sparse eigs) for spectral estimate ---
try:
    import scipy
    from scipy.sparse import csr_matrix
    from scipy.sparse.linalg import eigs
    HAVE_SCIPY = True
except Exception:
    subprocess.run([sys.executable, "-m", "pip", "install", "-q", "scipy"], check=False)
    try:
        import scipy
        from scipy.sparse import csr_matrix
        from scipy.sparse.linalg import eigs
        HAVE_SCIPY = True
    except Exception:
        HAVE_SCIPY = False
# ----- utils -----
def unitize(x, axis=-1, eps=1e-12):
    n = np.linalg.norm(x, axis=axis, keepdims=True) + eps
    return x / n
def sample_uniform_sphere(n, rng):
    x = rng.normal(size=(n, 3))
    return unitize(x)
def cap_area(eta):
    """Exact spherical cap area on S^2 (unit radius) for half-angle eta (rad)."""
    return 2.0 * math.pi * (1.0 - math.cos(eta))
# --- Analytic local term: g0 = 2 E[|x|^2] with uniform-in-cap area measure
# |x|^2 sin d = -|x|^2 cos + 2 sin + 2 cos + C
def g0_cap_analytic(eta):
    if eta <= 0:
        return 0.0
    I = -eta*eta*math.cos(eta) + 2*eta*math.sin(eta) + 2*(math.cos(eta) - 1.0)
    return 2.0 * I / (1.0 - math.cos(eta))
def tube_isotropic_ceiling_g(eta):
```



```

        """Memoryless upper bound:  $\min\{2^{-2}, 2 E^{[2]}_{\text{iso}}\}$ ,  $E^{[2]}_{\text{iso}} = (1/2) \sin^2 d$ ."""
        E_th2_iso = 2.934802200544679 # exact value of  $2^{-2} \sin^2 d$ 
        return min(2.0 * eta * eta, 2.0 * E_th2_iso)
def rho_cap_bound(eta):
    """Cap ceiling used here: average cos under uniform-in-cap area measure."""
    return 0.5 * (1.0 + math.cos(eta))
def spectral_rho_knn_nonlazy(U, k, ensure_connected=True, max_k_factor=4):
    """
    Estimate persistence from a k-NN graph on U (N,3) using NON-lazy random walk  $P = D^{-1}A$ .
    Ensures  $k \geq \text{ceil}(\log N)$ ; optionally bumps k to connect.
    Returns (rho_est, k_used). Falls back to dense eigendecomp if SciPy unavailable.
    """
    N = len(U)
    k = max(int(math.ceil(math.log(max(N, 3)))), int(k)) # ensure some mixing
    # cosine similarity (unit vectors)
    D = U @ U.T
    np.fill_diagonal(D, -np.inf)
    k_curr = max(2, min(N - 2, int(k)))
    while True:
        idx = np.argpartition(-D, kth=k_curr - 1, axis=1)[:k_curr]
        rows, cols, data = [], [], []
        for i in range(N):
            js = idx[i]
            rows.extend([i] * len(js)); cols.extend(js.tolist()); data.extend([1.0] * len(js))
            rows.extend(js.tolist()); cols.extend([i] * len(js)); data.extend([1.0] * len(js))
        if HAVE SCIPY:
            A = csr_matrix((data, (rows, cols)), shape=(N, N))
            A = A.minimum(A.T)
            deg = (A.sum(axis=1).A1)
            connected = np.all(deg > 0)
        else:
            A = np.zeros((N, N), float)
            A[rows, cols] = 1.0
            A = np.maximum(A, A.T)
            deg = A.sum(axis=1)
            connected = np.all(deg > 0)
        if not ensure_connected or connected:
            break
        k_new = min(N - 2, int(min(N - 2, max_k_factor * k)))
        if k_new == k_curr:
            break
        k_curr = k_new
    # NON-lazy random walk  $P = D^{-1}A$  (largest eigenvalue = 1)
    if HAVE SCIPY:
        deg = (A.sum(axis=1).A1) + 1e-12
        P = A.multiply(1.0 / deg[:, None])
        vals = eigs(P.T, k=2, which='LR', return_eigenvectors=False)
        vals = np.sort(np.real(vals))[:, -1]
        lam2 = float(vals[1]) if len(vals) > 1 else 0.0
    else:
        A = np.array(A)

```

```

        deg = A.sum(axis=1) + 1e-12
        P = (A / deg[:, None])
        vals = np.linalg.eigvals(P.T)
        vals = np.sort(np.real(vals))[:, -1]
        lam2 = float(vals[1]) if len(vals) > 1 else 0.0
        return max(0.0, min(0.9999, lam2)), k_curr
# ----- main fixed estimator + scans -----
def gshare_fixed(eta_rad, N=2048, mu_target=8, seed=11):
    """
    FIXED ESTIMATOR:
    - g0 from analytic cap-only expectation
    - from spectral kNN on next-shell nodes, CLIPPED by geometric cap bound
    - S = (1+)/(1-)
    """
    rng = np.random.default_rng(seed)
    U = sample_uniform_sphere(N, rng)
    k_init = max(2, int(round(mu_target)))
    rho_spec, k_used = spectral_rho_knn_nonlazy(U, k=k_init, ensure_connected=True)
    g0 = g0_cap_analytic(eta_rad)
    rho_clip = rho_cap_bound(eta_rad)
    rho = min(rho_spec, rho_clip)
    S = (1.0 + rho) / (1.0 - rho)
    gceil0 = tube_isotropic_ceiling_g(eta_rad)
    g_eff = S * g0
    gceil_eff = S * gceil0
    return {
        "N": N,
        "eta_deg": float(eta_rad * 180.0 / math.pi),
        "A_cap": float(cap_area(eta_rad)),
        "mu_target": mu_target,
        "g0_cap_only": g0,
        "rho_spectral_raw": rho_spec,
        "rho_cap_bound": rho_clip,
        "rho_used": rho,
        "S_multiplier": S,
        "g_share_estimate": g_eff,
        "g_ceiling_memoryless": gceil0,
        "g_ceiling_times_S": gceil_eff,
        "kNN_k_used": k_used,
        "sanity": {
            "violates_tube_ceiling_memoryless": bool(g0 > gceil0 + 1e-9),
            "violates_tube_ceiling_with_S": bool(g_eff > gceil_eff + 1e-9)
        }
    }
}

def scan_eta(start_deg=1.0, stop_deg=90.0, step_deg=0.5, **kwargs):
    out = []
    for d in np.arange(start_deg, stop_deg + 1e-9, step_deg):
        out.append(gshare_fixed(math.radians(d), **kwargs))
    return out

def find_eta_for_target(target_g=7.4, **kwargs):
    # brute search over 0.1° grid for stability

```

```

best = None
for d_tenths in range(10, 900): # 1.0° .. 90.0°
    eta = math.radians(d_tenths / 10.0)
    rep = gshare_fixed(eta, **kwargs)
    err = abs(rep["g_share_estimate"] - target_g)
    if (best is None) or (err < best["abs_error"]):
        rep2 = {**rep}
        rep2["target"] = target_g
        rep2["abs_error"] = err
        best = rep2
return best

# ----- run it -----
if __name__ == "__main__":
    seed = 11
    N = 2048
    mu_target = 8
    # 1) Hemisphere baseline (should be ~6.85)
    rep_hemi = gshare_fixed(math.radians(90.0), N=N, mu_target=mu_target, seed=seed)
    print("[hemisphere baseline]")
    print(json.dumps(rep_hemi, indent=2))
    # 2) Wide scan and report maximum in [1°, 90°]
    scan = scan_eta(start_deg=1.0, stop_deg=90.0, step_deg=1.0, N=N, mu_target=mu_target, seed=seed)
    best = max(scan, key=lambda r: r["g_share_estimate"])
    print("\n[ scan: max over 1°..90°]")
    print(json.dumps({k: best[k] for k in [
        "eta_deg", "g0_cap_only", "rho_cap_bound", "rho_used", "S_multiplier", "g_share_estimate"]},
        indent=2))
    # 3) Find closest to target g (default 7.4)
    near = find_eta_for_target(target_g=7.4, N=N, mu_target=mu_target, seed=seed)
    print("\n[closest to target g 7.4]")
    print(json.dumps({k: near[k] for k in [
        "eta_deg", "g0_cap_only", "rho_cap_bound", "rho_used", "S_multiplier",
        "g_share_estimate", "target", "abs_error"]}, indent=2))

```

Output (verbatim)

```

[hemisphere baseline]
{
  "N": 2048,
  "eta_deg": 90.0,
  "A_cap": 6.283185307179585,
  "mu_target": 8,
  "g0_cap_only": 2.283185307179587,
  "rho_spectral_raw": 0.9968026205673287,
  "rho_cap_bound": 0.5,
  "rho_used": 0.5,
  "S_multiplier": 3.0,
  "g_share_estimate": 6.849555921538761,
  "g_ceiling_memoryless": 4.934802200544679,
  "g_ceiling_times_S": 14.804406601634037,
  "kNN_k_used": 8,
  "sanity": {
    "violates_tube_ceiling_memoryless": false,

```

```

    "violates_tube_ceiling_with_S": false
  }
}
[ scan: max over 1°..90°]
{
  "eta_deg": 7.0,
  "g0_cap_only": 0.01492006179319609,
  "rho_cap_bound": 0.996273075820661,
  "rho_used": 0.996273075820661,
  "S_multiplier": 535.6355481786971,
  "g_share_estimate": 7.991715477458623
}
[closest to target g 7.4]
{
  "eta_deg": 61.9,
  "g0_cap_only": 1.1278039904207022,
  "rho_cap_bound": 0.735505940609705,
  "rho_used": 0.735505940609705,
  "S_multiplier": 6.561606504926233,
  "g_share_estimate": 7.400205999826242,
  "target": 7.4,
  "abs_error": 0.00020599982624158741
}

```